Lecture 2

Compactness, connectedness and gluing

Objectives. *Open sets let us define properties that distinguish spaces. Compactness expresses a finiteness condition on open covers. Connectedness rules out a separation into two open pieces. We prove how continuous maps preserve these properties, then use them to identify spaces formed by gluing. The main criterion turns a continuous bijection into a homeomorphism when its domain is compact and its target is Hausdorff.*

By the end of this lecture you will be able to:

- Prove the compact-to-Hausdorff criterion for a homeomorphism.
- Distinguish spaces by compactness, and by connectedness after removing a point.
- Test continuity out of a quotient and identify its points correctly.
- Describe the torus, Möbius band, projective plane and rotation group as quotients.

Unit 2A Compactness and connectedness

2.1 Compactness

Definition 2.1 (Compactness). An *open cover* of a space X is a family of open sets $(U_\alpha)_{\alpha\in I}$ whose union is X. A *subcover* is a subfamily that still covers X. The space is *compact* if every open cover has a finite subcover.

Compactness lets us replace data given on open sets near individual points by data on finitely many of those sets covering the whole space.

Definition 2.2 (Sequential compactness). A metric space is *sequentially compact* if every sequence in it has a subsequence converging to a point of that space.

This is the notion used in *Linear Algebra for Physics*, Lecture 13. It is equivalent to open-cover compactness for metric spaces (Munkres, Theorem 28.2). We use the open-cover definition on spaces without a specified metric.

A subset $K \subseteq X$ is called compact when its subspace topology is compact. Equivalently, every cover of K by open sets of X has a finite subcover. This follows because the open sets of K are intersections $K \cap U$ with ambient open sets U (§1.4).

The bounded interval $(0, 1)$ is not compact. The open sets

$$U_n = (1/n, 1), \qquad n \geq 2,$$

cover it, since every $x \in (0, 1)$ satisfies $1/n < x$ for some $n \geq 2$. A nonempty finite selection has a largest index N and misses $(0, 1/N]$. The empty selection covers no points. The figure compares this cover with a particular cover of $[0, 1]$.

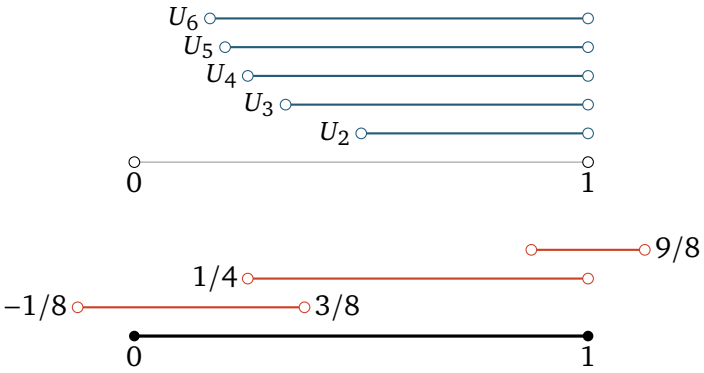


Figure 2.1: The sets $(1/n, 1)$ cover $(0, 1)$ with no finite subcover. The lower panel shows one finite cover of $[0, 1]$; compactness requires a finite subcover for every open cover.

Every finite space is compact. Choose one cover member containing each point. The resulting selection is finite. The empty space needs no members. In contrast, the real line is not compact. The open cover $\{(-n, n) : n \geq 1\}$ has no finite subcover, since a finite selection is contained in its member with the largest index.

Theorem 2.3 (Continuous images and closed subsets). *A continuous image of a compact space is compact. A closed subset of a compact space is compact.*

Proof. Let $f : X \to Y$ be continuous with X compact. Take an ambient open cover (V_α) of $f(X)$. Its inverse images cover X and are open. Choose finitely many of them covering X. The corresponding sets V_α cover $f(X)$.

Now let $C \subseteq X$ be closed, with X compact. Take an ambient open cover of C and add the open set $X \setminus C$. This covers X and therefore has a finite subcover. Discarding $X \setminus C$, if selected, leaves finitely many members covering C. ∎

We use two results about metric spaces with their proofs cited. *Heine–Borel* says that a subset of $\mathbb{R}^n$ is compact exactly when it is closed and bounded (Munkres, Theorem 27.3). Bounded means contained in some ball of finite radius. Thus closed intervals such as $[0, 1]$, closed balls and unit spheres are compact. The hypotheses refer to Euclidean space. Closed and bounded need not imply compact in a general metric space.

Compactness also gives extrema. For a continuous function $f : X \to \mathbb{R}$ on a nonempty compact space X, the image $f(X)$ is nonempty and compact. By Heine–Borel it is closed and bounded. Its supremum and infimum belong to its closure, hence to $f(X)$. Thus f attains a maximum and a minimum.

The circle is compact by Heine–Borel, whereas the real line is not. They cannot be homeomorphic. A homeomorphism would take the compact circle onto the line. This settles one distinction promised in Lecture 1.

2.2 Compact spaces and Hausdorff targets

Lecture 1 gave a continuous bijection with a discontinuous inverse. We seek hypotheses that force the inverse to be continuous. We follow closed sets: a closed subset of a compact domain is compact, and its continuous image is compact. We need that image to be closed. Hausdorffness supplies this last step. Recall that distinct points in a Hausdorff space have disjoint open neighbourhoods (§1.5).

In the figure, finitely many neighbourhoods cover the compact set K. Their corresponding neighbourhoods of y have an open intersection missing K.

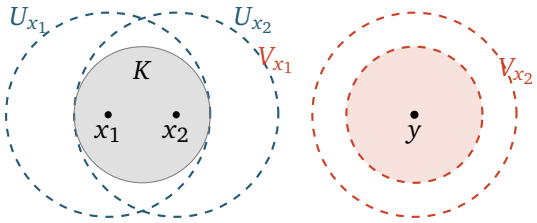


Figure 2.2: The blue neighbourhoods cover K. The red neighbourhoods of y each avoid the corresponding blue set, so their intersection avoids K.

Proposition 2.4 (Compact subsets are closed). *A compact subset of a Hausdorff space is closed.*

Proof. Let $K \subseteq Y$ be compact and $y \in Y \setminus K$. For each $x \in K$, choose disjoint open neighbourhoods U_x of x and V_x of y. Finitely many sets $U_{x_1}, \dots, U_{x_m}$ cover K. Finiteness matters here because finite intersections of open sets are open. The open neighbourhood $V_{x_1} \cap \cdots \cap V_{x_m}$ of y misses their union, so misses K. Thus $Y \setminus K$ is open. If K is empty, it is closed directly. ∎

Continuity can also be tested on closed sets. Inverse images commute with complements, so the open-set test is equivalent to the closed-set test.

Corollary 2.5 (Compact-to-Hausdorff criterion). *A continuous bijection from a compact space to a Hausdorff space is a homeomorphism.*

Proof. Let $f : X \to Y$ satisfy the hypotheses. For a closed subset $C \subseteq X$, compactness of X makes C compact. Its image $f(C)$ is compact, hence closed in Y. This image is the inverse image of C under $f^{-1} : Y \to X$. The closed-set test proves continuity of the inverse. ∎

Lecture 1's continuous bijection $[0, 2\pi) \to S^1$ does not meet the criterion. Its domain is not compact, since it is not closed in $\mathbb{R}$. We will instead use $[0, 1]$ and identify its two endpoints.

2.3 Connectedness

Definition 2.6 (Connectedness). A *separation* of a space X is a pair of disjoint, nonempty open sets whose union is X. The space is *connected* if it has no separation.

Definition 2.7 (Path). A *path* from x to y in X is a continuous map $\gamma : [0, 1] \to X$ with $\gamma(0) = x$ and $\gamma(1) = y$.

Definition 2.8 (Path-connectedness)**.** The space is *path-connected* if such a path exists for every pair $x, y \in X$.

A subset is connected or path-connected when its subspace topology has the corresponding property.

Theorem 2.9 (Intervals, images and paths)**.** *The interval* $[0,1]$ *is connected. Continuous images preserve connectedness and path-connectedness. Every path-connected space is connected.*

Proof. We argue by contradiction. A separation would put a supremum of points on one side into one of two open sets, and either choice contradicts that supremum. Suppose $[0,1] = U \cup V$ is a separation. Choose $a \in U$ and $b \in V$, with $a < b$ after exchanging the names if necessary. Put $c = \sup(U \cap [a,b])$. If $c \in U$, then $c < b$, and openness of U gives a point of $U \cap [a,b]$ larger than c, contradicting that c is an upper bound. If $c \in V$, then $c > a$, and openness of V forbids points of U immediately below c. This contradicts that points of $U \cap [a,b]$ approach their supremum from below. Both cases are impossible, so no separation exists.

For a continuous map $f\colon X \to Y$, a separation of $f(X)$ pulls back to a separation of X. Thus a connected domain has connected image. If X is path-connected, composing its paths with f gives paths between any two image points.

Finally, a path-connected space cannot have a separation $U \cup V$. A path joining a point of U to a point of V would have connected image, since $[0,1]$ is connected. Its intersections with U and V would separate that image. ∎

For example, a square and a ball are path-connected. The segment $\gamma(t) = (1-t)x + ty$ stays in the set and joins any two of its points.

Definition 2.10 (Connected component)**.** A *connected component* is a connected subset maximal under inclusion.

The component of $x \in X$ is the union of all connected subsets containing x. This union is connected. In any proposed separation, each of its connected subsets must lie wholly on the side containing x. It is maximal by its definition. Two components that meet therefore coincide, so components partition X.

Removing a point distinguishes a line from a plane, as the figure shows.

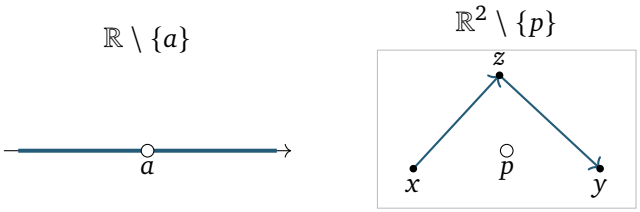


Figure 2.3: Removing a point separates a line into two open rays. In a plane, a broken line can join two points while avoiding the removed point.

The space $\mathbb{R} \setminus \{a\}$ has a separation into its two open rays. In $\mathbb{R}^n \setminus \{p\}$, with $n \geq 2$, take points x and y and choose z outside the two lines through p, x and through p, y. The segments from x to z and from z to y avoid p. They give the path

$$\gamma(t) = \begin{cases} (1-2t)x + 2tz, & 0 \leq t \leq \frac{1}{2}, \\ (2-2t)z + (2t-1)y, & \frac{1}{2} \leq t \leq 1. \end{cases}$$

The two formulas agree at $t = 1/2$. Each affine formula is continuous, so the preimage of a closed set is closed on its half-interval. It is also closed in $[0, 1]$, since both half-intervals are closed there. Their finite union is closed, proving continuity of γ. Thus the punctured space is path-connected.

We can now distinguish the line from the plane, as proposed in Lecture 1. Suppose $f : \mathbb{R} \to \mathbb{R}^2$ were a homeomorphism. Its restriction would give a homeomorphism

$$\mathbb{R} \setminus \{0\} \longrightarrow \mathbb{R}^2 \setminus \{f(0)\}.$$

The domain is disconnected and the target is path-connected, hence connected. A homeomorphism preserves connectedness, so $\mathbb{R} \not\equiv \mathbb{R}^2$. Removing a point cannot distinguish $\mathbb{R}^2$ from $\mathbb{R}^3$, since both remain path-connected. Lecture 3 introduces loops and their continuous deformations to make finer distinctions.

The radial map $x \mapsto x/\|x\|$ also shows that S^n is path-connected for $n \geq 1$, as an image of $\mathbb{R}^{n+1} \setminus \{0\}$.

Remark 2.11 (Optional: topologist's sine curve)**.** The closure of $\{(x, \sin(1/x)) : 0 < x \leq 1\}$ in $\mathbb{R}^2$ is connected but not path-connected (cited: Munkres, §24, Example 7). Later arguments do not depend on this example.

The figure shows part of the graph and the segment added by taking its closure.

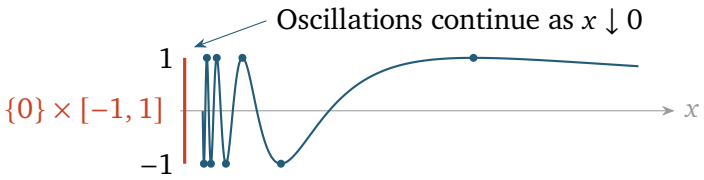


Figure 2.4: The plotted graph has $1/(8\pi) \leq x \leq 1$; its oscillations continue towards the added vertical segment. The marked extrema have heights 1 and -1.

Self-check. Prove that S^1 and $[0, 1]$ are not homeomorphic by removing one point. Explain why deleting an endpoint of the interval is not a useful test.

Answer. Delete the midpoint of $[0, 1]$. The two resulting half-intervals give a separation. Deleting any point from the circle leaves a space homeomorphic to $\mathbb{R}$. Rotate that point to the north pole, then use Lecture 1's stereographic projection. The line is path-connected by straight segments. A homeomorphism would restrict to a homeomorphism between these two punctured spaces, which is impossible. Deleting an endpoint leaves a path-connected half-open interval, so connectedness would not distinguish them.

Unit 2B Gluing

2.4 Quotients and orbit spaces

A periodic coordinate identifies the two ends of an interval. For $[0,1]$ with $0 \sim 1$, a neighbourhood of the identified point must include points near both ends. For $0 < \varepsilon < 1/2$, the set

$$[0,\varepsilon) \cup (1-\varepsilon, 1]$$

is open in $[0,1]$. We want it to be the inverse image of an open neighbourhood after gluing. We define openness in the quotient by this inverse-image rule.

Definition 2.12 (Quotient topology). For an equivalence relation $\sim$ on X, the *quotient set* $X/\!\sim$ consists of the equivalence classes. The projection $q\colon X \to X/\!\sim$ sends each point to its class. The *quotient topology* is defined by

$$V \subseteq X/\!\sim \text{ is open} \quad\Longleftrightarrow\quad q^{-1}(V) \text{ is open in } X.$$

Inverse images preserve unions and finite intersections, so this is a topology. It is the largest topology on the quotient set that makes the projection continuous. The projection is also surjective. Hence a quotient of a compact or path-connected space retains that property, as a continuous image (Theorems 2.3 and 2.9).

Proposition 2.13 (Continuity out of a quotient). *A map $\bar{f}\colon X/\!\sim \to Y$ is continuous exactly when $\bar{f}\circ q$ is continuous. A map $f\colon X \to Y$ that is constant on equivalence classes induces a unique map $\bar{f}$ with $f = \bar{f}\circ q$.*

Proof. Composition with continuous q preserves continuity. Conversely, if $\bar{f}\circ q$ is continuous, then for open $W \subseteq Y$ the set $q^{-1}(\bar{f}^{-1}(W))$ is open. The quotient definition says that $\bar{f}^{-1}(W)$ is open. The induced map is given by $\bar{f}([x]) = f(x)$, which is well-defined precisely when f is constant on each class. Surjectivity of q makes it unique. ∎

Definition 2.14 (Fibre). A *fibre* is the inverse image of one point.

Method for identifying a quotient. To identify $X/\!\sim$ with Y, we proceed as follows.

1. Construct a continuous surjection $f\colon X \to Y$.
2. Prove that its fibres are exactly the equivalence classes.
3. Prove that X is compact and Y is Hausdorff.

The quotient is compact as an image of X. The induced map is a continuous bijection by Proposition 2.13, so Corollary 2.5 makes it a homeomorphism.

Define the circle map $c\colon [0,1] \to S^1$ by

$$c(t) = (\cos(2\pi t), \sin(2\pi t)).$$

The figure depicts its endpoint identification and the inverse image of an open arc through the identified point, with $\varepsilon = 1/8$.

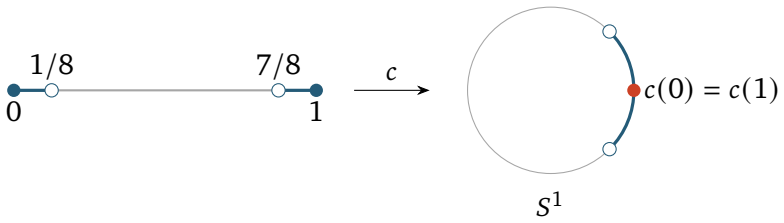


Figure 2.5: The coloured open arc has inverse image $[0, 1/8) \cup (7/8, 1]$, open in $[0, 1]$. The filled interval endpoints have the same image under c.

This map is continuous and surjective onto S^1. Its only repeated value on $[0, 1]$ is the common endpoint value. The induced map $[0, 1]/(0 \sim 1) \to S^1$ is therefore a continuous bijection. Its domain is compact and its target Hausdorff, so it is a homeomorphism by Corollary 2.5.

Group actions describe quotients by symmetries. We first define a group.

Definition 2.15 (Group). A *group* is a set G with an associative product $(g, h) \mapsto gh$, an *identity* $e \in G$ with $eg = ge = g$ for every $g \in G$, and, for each $g \in G$, an *inverse* $g^{-1} \in G$ with $gg^{-1} = g^{-1}g = e$.

Definition 2.16 (Group action). An *action by homeomorphisms* of a group G on X assigns a homeomorphism $x \mapsto gx$ to each $g \in G$, with $ex = x$ and $(gh)x = g(hx)$.

Definition 2.17 (Orbit). The *orbit* of x is $\{gx : g \in G\}$.

Definition 2.18 (Orbit space). The *orbit space* X/G has the quotient topology for these orbits.

For example, the additive group $\mathbb{Z}$ acts on $\mathbb{R}$ by integer translations. The orbit of $x \in \mathbb{R}$ is $\{x + n : n \in \mathbb{Z}\}$. Every orbit meets $[0, 1]$. The only distinct points of $[0, 1]$ in the same orbit are its two endpoints. The figure marks one orbit and its representative in $[0, 1]$.

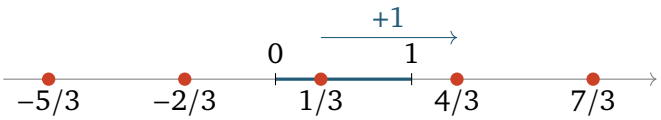


Figure 2.6: The orbit of $1/3$ consists of its integer translates. Exactly one of these points lies in $[0, 1]$.

We prove Hausdorffness for the projective and Möbius quotients by constructing a metric on their orbits. We first verify the metric axioms. We then show that its open sets are exactly the quotient-open sets.

Definition 2.19 (Isometry). An *isometry* preserves a metric.

Suppose $a\colon X \to X$ is an isometry with $a^2 = \mathrm{id}$. Its orbit classes are $[x] = \{x, ax\}$, with duplicates omitted. The orbit distance is

$$d_{\mathrm{orb}}([x], [y]) = \min\{d(x, y), d(x, ay)\}.$$

The isometry and $a^2 = \mathrm{id}$ give $d(ax, y) = d(a^2x, ay) = d(x, ay)$. Thus changing either

representative interchanges the two distances. The minimum vanishes exactly when the orbits agree. Symmetry follows from the isometry.

For the triangle inequality, choose maps $g, h \in \{\mathrm{id}, a\}$ such that

$$d_{\mathrm{orb}}([x],[y]) = d(x, gy), \qquad d_{\mathrm{orb}}([y],[z]) = d(y, hz).$$

The composition gh is again either id or a. By the triangle inequality and the fact that g is an isometry,

$$d(x, ghz) \le d(x, gy) + d(gy, ghz) = d(x, gy) + d(y, hz).$$

This bounds the orbit distance from $[x]$ to $[z]$ by the sum of the other two. Thus the formula defines a metric.

For $r > 0$, its ball centred at $[x]$ is $q(B_r(x))$: a class lies in that ball exactly when one of its representatives lies in $B_r(x)$. The inverse image of this ball is $B_r(x) \cup B_r(ax)$. Thus orbit-metric balls are quotient-open. Conversely, the preimage of a quotient-open set containing $[x]$ contains some $B_r(x)$. Its image is an orbit-metric ball inside that set, so the set is metric-open. The topologies agree. The orbit space is therefore Hausdorff by Proposition 1.25.

Definition 2.20 (Real projective space)**.** For $n \ge 1$, the *real projective space* $\mathbb{R}\mathrm{P}^n$ is $S^n/(x \sim -x)$. Its points are the lines through the origin in $\mathbb{R}^{n+1}$.

The antipodal map is an isometry, so $\mathbb{R}\mathrm{P}^n$ is Hausdorff. It is compact and path-connected as an image of S^n. Write $D^n = \{v \in \mathbb{R}^n : \|v\| \le 1\}$ for the closed unit disc. Each line meets S^n in an antipodal pair. Keeping the upper hemisphere retains exactly one point of each pair off the equator. On the equator both points remain. Projection onto the first n coordinates identifies the upper hemisphere with a closed disc. Thus only boundary points of that disc still have partners. The figure shows this construction for $n = 2$. The blue point $p = (3/5, 0, 4/5)$ projects to $v = (3/5, 0)$ in the disc. The red points $b = (1, 0, 0)$ and $-b$ remain paired.

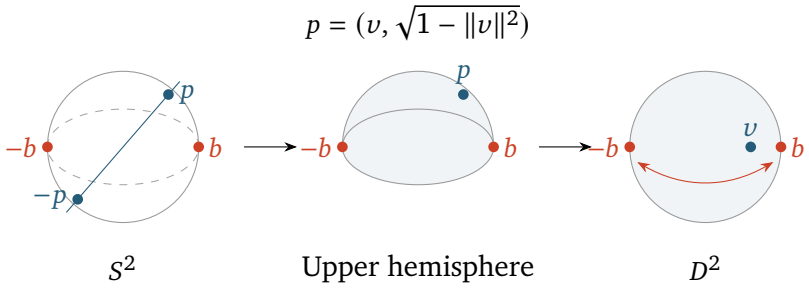


Figure 2.7: The upper hemisphere retains one representative of a non-equatorial pair and both representatives of an equatorial pair. Projection to D^2 leaves only boundary antipodes paired; the same construction gives D^3 for $\mathbb{R}\mathrm{P}^3$.

The map

$$D^n \longrightarrow \mathbb{R}\mathrm{P}^n, \qquad v \longmapsto [(v, \sqrt{1 - \|v\|^2})]$$

is continuous and surjective. The upper hemisphere consists of unit vectors whose last coordinate is nonnegative. Every line has a representative there. That representative is unique unless its last coordinate vanishes, when there are two antipodal representatives. Thus this realizes $\mathbb{R}\mathrm{P}^n$ as D^n with antipodal points of its boundary sphere identified, again by the compact-to-Hausdorff criterion.

The Hausdorff conclusion fails for general orbit spaces. Let the additive group $\mathbb{Q}$ act on $\mathbb{R}$ by translation. A nonempty open union U of orbits contains an interval. For any $x \in \mathbb{R}$, density of the rationals supplies $r \in \mathbb{Q}$ with $x - r$ in that interval. Invariance under translation by r gives $x \in U$. Hence the only open sets of $\mathbb{R}/\mathbb{Q}$ are the empty set and the whole space. There are distinct classes, for example those of 0 and $\sqrt{2}$, so this quotient is not Hausdorff. The Hausdorff condition will matter for the spaces constructed in Lectures 7 and 8.

2.5 Rotations as a glued ball

Definition 2.21 (Rotation group)**.** The rotation group is the matrix set

$$\mathrm{SO}(3) = \{R \in \mathbb{R}^{3\times 3} : R^T R = \mathbb{1},\ \det R = 1\},$$

with its Euclidean subspace topology.

A vector v with $\|v\| \le \pi$ will specify an axis and an angle. The figure shows the identification that occurs at angle π, a half-turn. The vector n marks a unit axis.

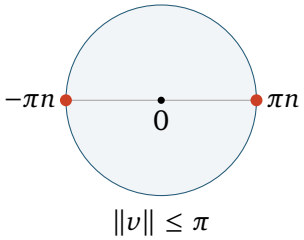


Figure 2.8: A cross-section of the closed rotation ball. Interior vectors are distinct rotations. The matching red boundary points πn and $-\pi n$ represent the same half-turn.

Proposition 2.22 (Axis–angle ball)**.** *The closed ball* $\{v \in \mathbb{R}^3 : \|v\| \le \pi\}$ *with antipodal boundary points identified is homeomorphic to* $\mathrm{SO}(3)$*. Consequently,* $\mathrm{SO}(3) \cong \mathbb{R}\mathrm{P}^3$.

Proof. We construct a continuous rotation map from the ball, prove it is onto, and determine its fibres. The compact-to-Hausdorff criterion then identifies the quotient.

For $v = rn$ with $r > 0$ and $\|n\| = 1$, split a vector $w \in \mathbb{R}^3$ into its axis and perpendicular parts:

$$w = (n \cdot w)n + w_\perp.$$

A rotation around n fixes the axis part. On the perpendicular plane it sends $w_\perp$ to $(\cos r)w_\perp + (\sin r)n \times w_\perp$. The figure shows these parts for $n = (0, 0, 1)$, $w = (1, 0, 1)$ and $r = \pi/3$.

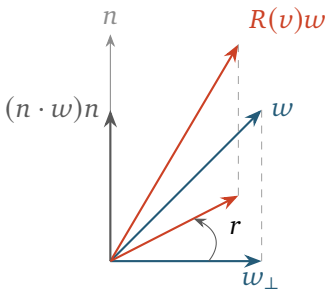


Figure 2.9: The axis part remains fixed. The perpendicular part rotates through $\pi/3$, sending $w = (1, 0, 1)$ to $(1/2, \sqrt{3}/2, 1)$.

The resulting axis–angle formula is

$$R(v)w = (\cos r)w + (1 - \cos r)n(n \cdot w) + (\sin r)n \times w.$$

We need a formula continuous at $v = 0$, where n is undefined. Define $A_v(w) = v \times w$ for $w \in \mathbb{R}^3$. Put $r = \|v\|$ and define

$$R(v) = \mathbb{1} + \frac{\sin r}{r} A_v + \frac{1 - \cos r}{r^2} A_v^2.$$

At $r = 0$, the coefficients have limiting values 1 and $1/2$. The map is therefore continuous there and $R(0) = \mathbb{1}$. For $v = rn$ with $\|n\| = 1$, the vector identity $n \times (n \times w) = n(n \cdot w) - w$ recovers the axis–angle formula. This fixes n and rotates its perpendicular plane through angle r. Thus it is orthogonal with determinant 1. At $r = \pi$ it equals $2nn^T - \mathbb{1}$, unchanged by $n \mapsto -n$.

Conversely, every $R \in \mathrm{SO}(3)$ has a unit fixed vector n. Indeed, orthogonality forces all eigenvalues to have modulus 1. Its nonreal eigenvalues occur in conjugate pairs of product 1. Its real eigenvalues are ± 1. After removing the conjugate pairs, an odd number of real eigenvalues remains. Their product is 1, so an even number are -1. At least one is therefore $+1$.

The perpendicular plane is invariant. Since $Rn = n$ and R is orthogonal, for $w \perp n$ we have $Rw \cdot n = Rw \cdot Rn = w \cdot n = 0$. The restriction has determinant 1, so is a plane rotation. Choosing its angle in $[-\pi, \pi]$ represents R as $R(\theta n)$.

There are no further identifications. The trace is $1 + 2\cos r$, so it determines $r \in [0, \pi]$. For $0 < r < \pi$, the difference $R(v) - R(v)^T = 2 \sin r \, A_n$ determines n. At $r = \pi$, the matrix $2nn^T - \mathbb{1}$ determines the line spanned by n, hence just the pair $\{n, -n\}$. At $r = 0$ there is only the vector 0.

We have a continuous bijection from the compact quotient ball to the Hausdorff matrix space $\mathrm{SO}(3)$. It is a homeomorphism. Rescaling the ball to D^3 and using the hemisphere construction above identifies it with $\mathbb{R}\mathrm{P}^3$. ∎

Lecture 4 will use this model to distinguish loops representing full rotations. No property of loops is needed for the identification here.

2.6 Three quotients of a square

We determine compactness, path-connectedness and Hausdorffness for the spaces obtained by the edge pairings in the figure. All start from $Q_0 = [0, 1]^2$. Matching arrows indicate which points are identified. Apply also all identifications forced transitively by these pairings.

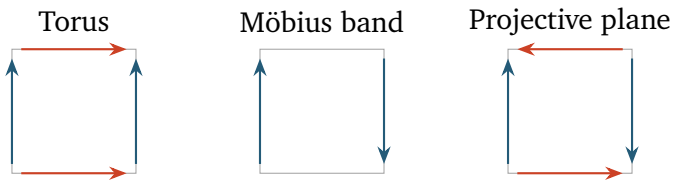


Figure 2.10: The torus pairs opposite edges without reversal. The Möbius band pairs only the vertical edges, with reversal. The projective plane pairs antipodal points of the square boundary.

Each quotient is compact and path-connected, since the square has both properties and the projection is continuous. Hausdorffness needs a separate argument. It need not survive a quotient, as $\mathbb{R}/\mathbb{Q}$ showed.

The following figure shows the surfaces described by the edge pairings. The torus closes both pairs of edges. The Möbius band closes one pair with a half-twist and retains one boundary component. In the cylinder panel, z is a unit complex number and $u \in [-1, 1]$ is the height.

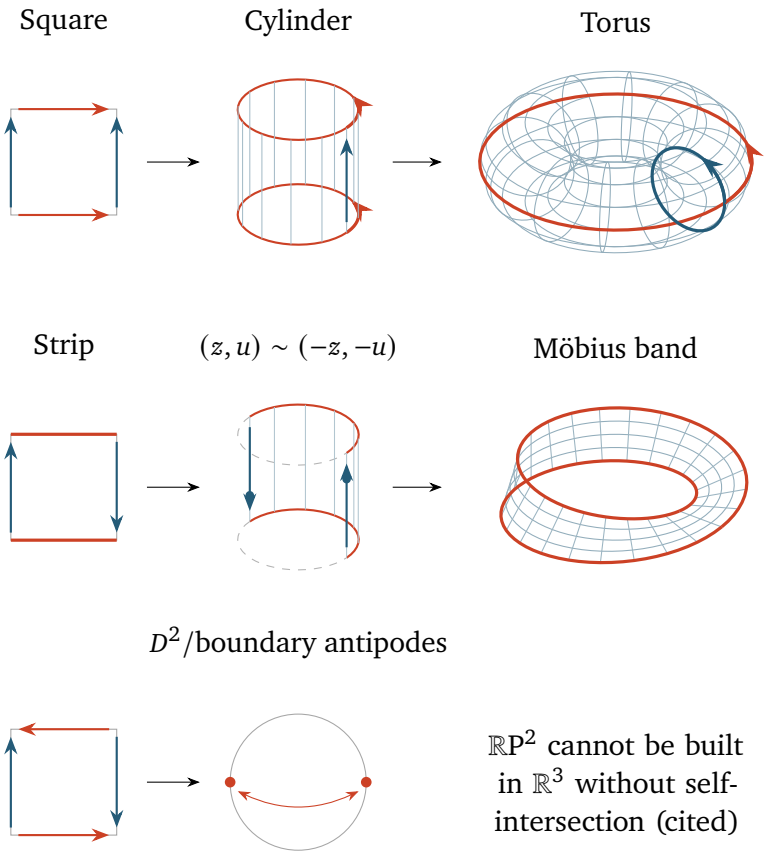


Figure 2.11: The square gives a cylinder and then a torus; the strip with reversed ends gives a Möbius band with one boundary component. The projective-plane row shows identifications; its non-embedding statement is cited from Hatcher, *Algebraic Topology*, Corollary 3.46, p. 256.

For the *torus*, the pairings are

$$(0, t) \sim (1, t) \quad \text{and} \quad (s, 0) \sim (s, 1).$$

Use the circle map c from §2.4.

Definition 2.23 (Torus). The product $T^2 = S^1 \times S^1 \subseteq \mathbb{R}^4$ is the torus.

The map $Q_0 \to T^2$ given by $(s,t) \mapsto (c(s), c(t))$ is continuous and onto. The familiar torus of revolution is the image of Q_0 under the map to $\mathbb{R}^3$ given by

$$\big((2+\cos(2\pi t))\cos(2\pi s),\ (2+\cos(2\pi t))\sin(2\pi s),\ \sin(2\pi t)\big).$$

Its distance from the vertical axis is $2+\cos(2\pi t) > 0$. This distance and the height recover $c(t)$, and the horizontal direction recovers $c(s)$. Hence its fibres are the same as those of $(c(s), c(t))$. The compact-to-Hausdorff criterion identifies the product torus with this surface.

Equality in the product is coordinatewise. In each coordinate, c identifies only the two endpoints. Thus the map's fibres are exactly the stated equivalence classes, including the single class of all four corners. The induced continuous bijection is a homeomorphism, because the quotient is compact and T^2 is Hausdorff.

Definition 2.24 (Möbius band). For the *Möbius band*, the pairing is $(0,t) \sim (1, 1-t)$.

We construct the half-twist from a cylinder by keeping its upper semicircle. This gives a rectangle of representatives. Identifying the two ends of that semicircle must reverse the height coordinate to give the stated pairing. Identify S^1 with unit complex numbers. On the cylinder $S^1 \times [-1,1] \subseteq \mathbb{R}^3$, the map $(z,u) \mapsto (-z,-u)$ is an isometry whose square is the identity. Its orbit space is therefore Hausdorff by the orbit metric above. The map

$$Q_0 \longrightarrow (S^1 \times [-1,1])/\{\pm 1\}, \qquad (s,t) \longmapsto [(e^{i\pi s}, 2t-1)]$$

is continuous and onto. Every circle point or its negative lies on the upper semicircle.

For $0 < s < 1$ that representative is unique. At the two ends, changing 1 to -1 reverses the other coordinate, giving precisely the stated edge pairing. The induced bijection is a homeomorphism by the same compact-to-Hausdorff criterion.

For the *projective plane*, the pairings are

$$(0,t) \sim (1, 1-t) \quad \text{and} \quad (s,0) \sim (1-s, 1).$$

The radial map F in the next figure sends concentric squares to concentric circles. Each point stays on its ray, so antipodes remain antipodes.

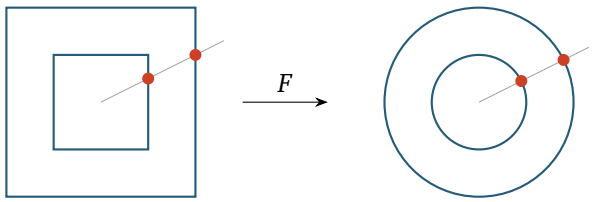


Figure 2.12: The contours $\|w\|_\infty = 1/2$ and $\|w\|_\infty = 1$ map to circles of radii $1/2$ and 1. The marked points lie on the ray through $(1, 1/2)$.

Put $w = (2s-1, 2t-1)$. The centred square maps homeomorphically to D^2 by radial rescaling:

$$w \longmapsto \frac{\|w\|_\infty}{\|w\|} w \quad (w \neq 0), \qquad 0 \longmapsto 0,$$

where $\|w\|_\infty$ is the maximum absolute coordinate and $\|w\|$ is Euclidean. The inverse sends nonzero u to $\|u\|\ u/\|u\|_\infty$. Their norms tend to zero at the origin, so both maps are continuous there as well. The square boundary maps to the unit circle and antipodes map to antipodes. The hemisphere construction therefore identifies this square quotient with $\mathbb{R}\mathrm{P}^2$, whose Hausdorffness was proved using its orbit metric.

The results, including the rotation ball, are as follows.

Quotient	Compact and path-connected	Hausdorff witness
Torus	yes	subspace of $\mathbb{R}^4$
Möbius band	yes	isometric orbit space
Projective plane	yes	antipodal orbit metric
Rotation ball	yes	$\mathrm{SO}(3) \subseteq \mathbb{R}^9$

Self-check. Let Q_0 have the torus edge pairings. Determine whether the continuous surjection $f\colon Q_0 \to S^1 \times S^1$ given by

$$f(s,t) = (c(s+2t), c(3t))$$

induces a homeomorphism from that quotient. Here the same formula for c is defined on all of $\mathbb{R}$. Compare the classes of $(0,0)$ and $(1/3,1/3)$, and identify which condition of the quotient method decides the answer.

Answer. Changing s or t by 1 leaves f unchanged, so it descends to the quotient. It is onto: choose $t \in [0,1/3]$ for the second circle coordinate, then choose $s \in [0,1]$ for the first. But

$$f(0,0) = (c(0), c(0)) = f(1/3,1/3),$$

since both arguments of c at $(1/3,1/3)$ equal 1. The interior point $(1/3,1/3)$ is not equivalent to a corner. Thus a fibre contains two different equivalence classes. The induced map is not injective, so it is not a homeomorphism. The map $(s,t) \mapsto (c(s), c(t))$ has exactly the required fibres instead.

Self-check. State in one sentence what the quotient topology is and why we introduce it. Explain what its rule requires at the glued point of $[0,1]/(0 \sim 1)$.

Answer. The quotient topology makes a set of equivalence classes open exactly when its inverse image under the projection is open, so we can construct spaces by gluing and test continuous maps out of them before gluing. At the glued point an open inverse image contains both endpoints, hence contains $[0,\varepsilon) \cup (1-\varepsilon, 1]$ for some $0 < \varepsilon < 1/2$. A set whose inverse image contains only a neighbourhood of 0 cannot be a neighbourhood of the glued point, since that inverse image must also contain 1.

Summary. Continuous images preserve compactness and connectedness. A continuous bijection from compact to Hausdorff is a homeomorphism. A quotient topology tests continuity after composition with its projection. Quotients preserve compactness and path-connectedness, but Hausdorffness needs its own proof. The rotation group is the closed three-ball with antipodal boundary points identified, hence $\mathbb{R}\mathrm{P}^3$. Lecture 3 begins the study of loops.

Reading. Munkres, *Topology*, second edition, §§22–28; Heine–Borel is Theorem 27.3, metric sequential compactness is Theorem 28.2, and the sine curve is §24, Example 7.